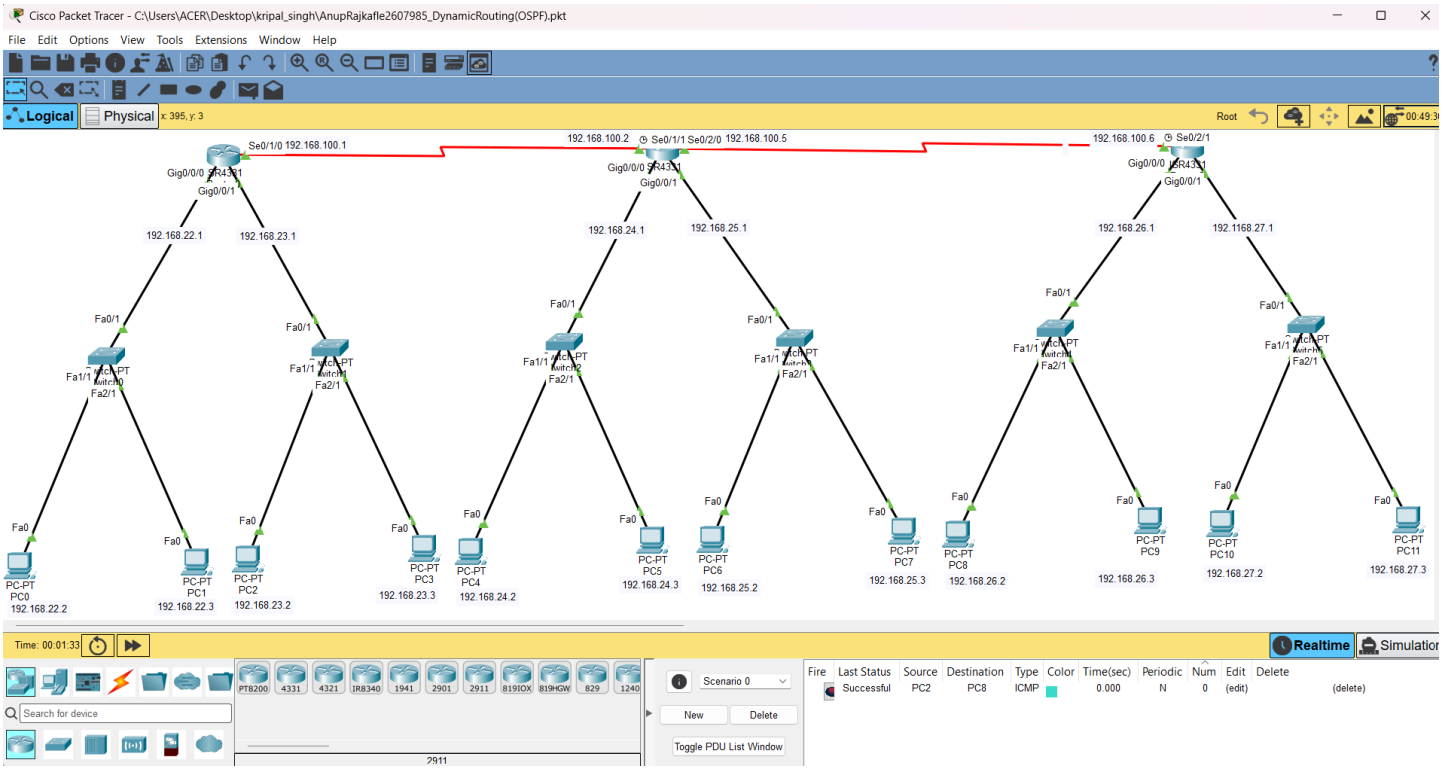
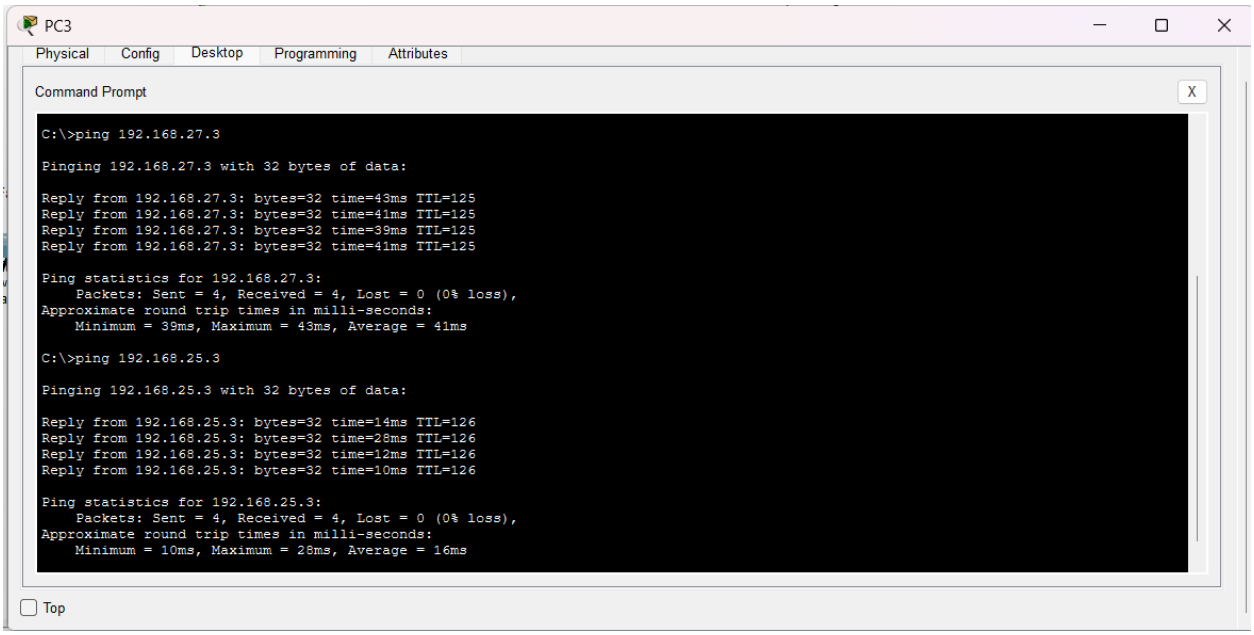
Fully configured network topology

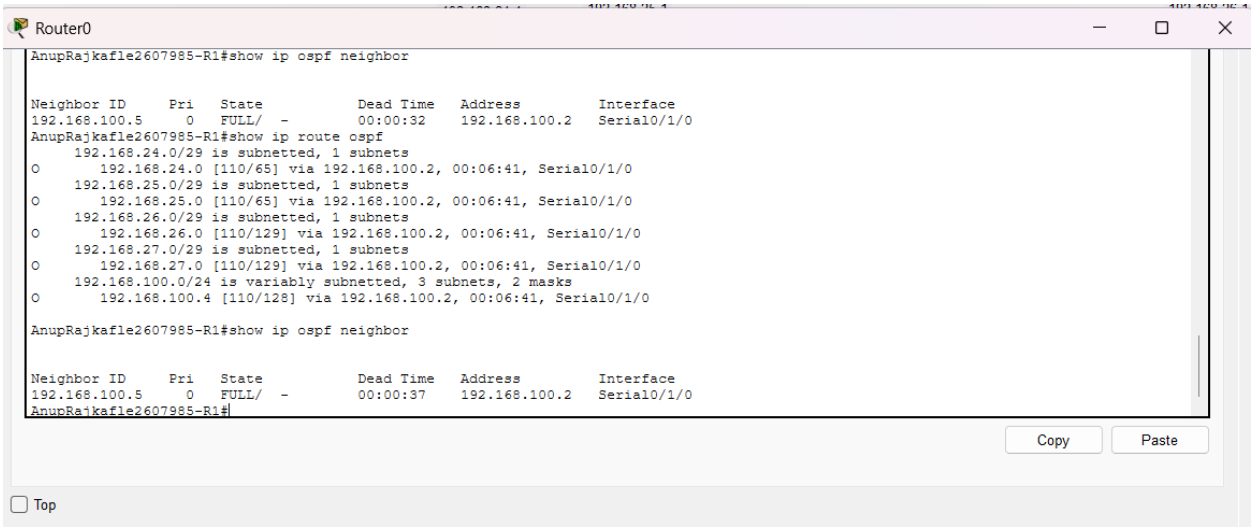


Verification: ping between different pc

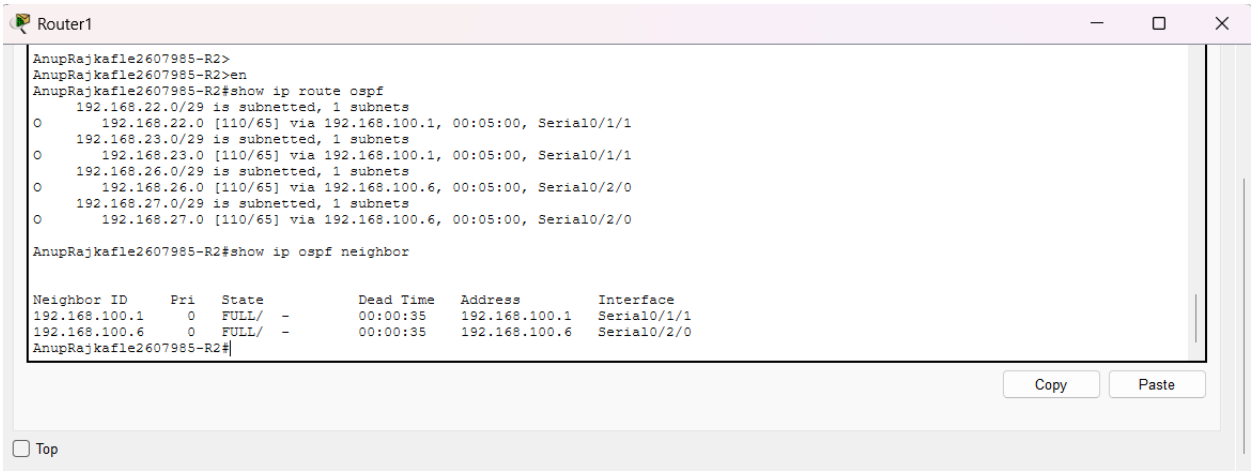


IP route OSPF & OSPF neighbor

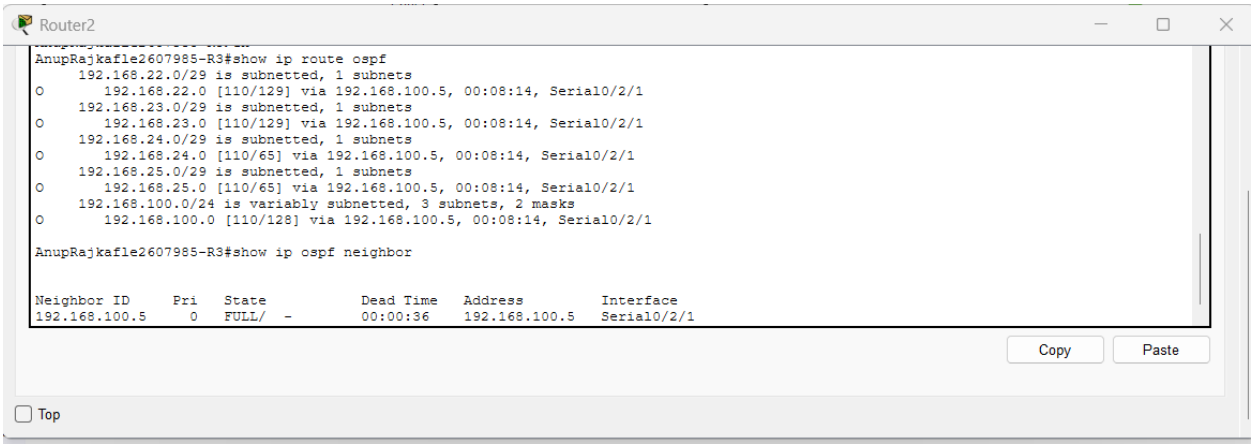
AnupRajkafle2607985-R1



AnupRajkafle2607985-R2



AnupRajkafle2607985-R3



IP addressing scheme with Appropriate CIDR addressing

Device	Interface / Connection	Assigned IP Address	CIDR Prefix	Subnet Mask
AnupRajkafle2607985-R1	Serial0/1/0	192.168.100.1	/30	255.255.255.252
	GigabitEthernet0/0/0	192.168.22.1	/29	255.255.255.248
	GigabitEthernet0/0/1	192.168.23.1	/29	255.255.255.248
AnupRajkafle2607985-R2	Serial0/1/1	192.168.100.2	/30	255.255.255.252
	Serial0/2/0	192.168.100.5	/30	255.255.255.252
	GigabitEthernet0/0/0	192.168.24.1	/29	255.255.255.248
	GigabitEthernet0/0/1	192.168.25.1	/29	255.255.255.248
AnupRajkafle2607985-R3	Serial0/2/1	192.168.100.6	/30	255.255.255.252
	GigabitEthernet0/0/0	192.168.26.1	/29	255.255.255.248

Device	Interface / Connection	Assigned IP Address	CIDR Prefix	Subnet Mask
	GigabitEthernet0/0/1	192.168.27.1	/29	255.255.255.248
PC0	FastEthernet0	192.168.22.2	/29	255.255.255.248
PC1	FastEthernet0	192.168.22.3	/29	255.255.255.248
PC2	FastEthernet0	192.168.23.2	/29	255.255.255.248
PC3	FastEthernet0	192.168.23.3	/29	255.255.255.248
PC4	FastEthernet0	192.168.24.2	/29	255.255.255.248
PC5	FastEthernet0	192.168.24.3	/29	255.255.255.248
PC6	FastEthernet0	192.168.25.2	/29	255.255.255.248
PC7	FastEthernet0	192.168.25.3	/29	255.255.255.248

Device	Interface / Connection	Assigned IP Address	CIDR Prefix	Subnet Mask
PC8	FastEthernet0	192.168.26.2	/29	255.255.255.248
PC9	FastEthernet0	192.168.26.3	/29	255.255.255.248
PC10	FastEthernet0	192.168.27.2	/29	255.255.255.248
PC11	FastEthernet0	192.168.27.3	/29	255.255.255.248

Command Export File

Thu Sep 3 21:53:22 2026 Router0 Router>en

Thu Sep 3 21:53:26 2026 Router0 Router#config t

Thu Sep 3 21:53:43 2026 Router0 Router(config)#hostname AnupRajkafle2607985-R1

Thu Sep 3 21:54:08 2026 Router1 Router>EN

Thu Sep 3 21:54:17 2026 Router1 Router#CONFIG T

Thu Sep 3 21:54:32 2026 Router1 Router(config)#HOSTNAME AnupRajkafle2607985-R2

Thu Sep 3 21:54:50 2026 Router2 Router>EN

Thu Sep 3 21:54:54 2026 Router2 Router#CONFIG T

Thu Sep 3 21:55:08 2026 Router2 Router(config)#HOSTNAME AnupRajkafle2607985-R3

Thu Sep 3 21:55:43 2026 Router0 AnupRajkafle2607985-R1(config)#INTERFACE se0/1/0

Thu Sep 3 21:58:19 2026 Router0 AnupRajkafle2607985-R1(config-if)#ip address 192.168.100.1 255.255.255.252

Thu Sep 3 21:58:33 2026 Router0 AnupRajkafle2607985-R1(config-if)#no shutdown

Thu Sep 3 21:58:37 2026 Router0 AnupRajkafle2607985-R1(config-if)#exit

Thu Sep 3 21:58:59 2026 Router0 AnupRajkafle2607985-R1(config)#interface gig0/0/0

Thu Sep 3 21:59:44 2026 Router0 AnupRajkafle2607985-R1(config-if)#ip address 192.168.22.1 255.255.255.248

Thu Sep 3 21:59:48 2026 Router0 AnupRajkafle2607985-R1(config-if)#no shut

Thu Sep 3 21:59:54 2026 Router0 AnupRajkafle2607985-R1(config-if)#exit

Thu Sep 3 22:00:10 2026 Router0 AnupRajkafle2607985-R1(config)#interface gig0/0/1

Thu Sep 3 22:00:46 2026 Router0 AnupRajkafle2607985-R1(config-if)#ip address 192.168.23.1 255.255.255.248

Thu Sep 3 22:00:48 2026 Router0 AnupRajkafle2607985-R1(config-if)#no shut

Thu Sep 3 22:00:54 2026 Router0 AnupRajkafle2607985-R1(config-if)#exit

Thu Sep 3 22:01:56 2026 Router1 AnupRajkafle2607985-R2>en

Thu Sep 3 22:01:59 2026 Router1 AnupRajkafle2607985-R2#config t

Thu Sep 3 22:02:58 2026 Router1 AnupRajkafle2607985-R2(config)#interface se0/1/1

Thu Sep 3 22:03:42 2026 Router1 AnupRajkafle2607985-R2(config-if)#ip address 192.168.100.2 255.255.255.252

Thu Sep 3 22:03:45 2026 Router1 AnupRajkafle2607985-R2(config-if)#no shut

Thu Sep 3 22:03:49 2026 Router1 AnupRajkafle2607985-R2(config-if)#exit

Thu Sep 3 22:04:12 2026 Router1 AnupRajkafle2607985-R2(config)#interface

Thu Sep 3 22:04:26 2026 Router1 AnupRajkafle2607985-R2(config)#interface gig0/0/0

Thu Sep 3 22:08:07 2026 Router1 AnupRajkafle2607985-R2(config-if)#ip address 192.168.24.1 255.255.255.248

Thu Sep 3 22:08:11 2026 Router1 AnupRajkafle2607985-R2(config-if)#exit

Thu Sep 3 22:08:42 2026 Router1 AnupRajkafle2607985-R2(config)#interface gig0/0/1

Thu Sep 3 22:09:23 2026 Router1 AnupRajkafle2607985-R2(config-if)#exit

Thu Sep 3 22:09:41 2026 Router1 AnupRajkafle2607985-R2(config)#interface gig0/0/0

Thu Sep 3 22:09:51 2026 Router1 AnupRajkafle2607985-R2(config-if)#ip address 192.168.24.1 255.255.255.248

Thu Sep 3 22:09:54 2026 Router1 AnupRajkafle2607985-R2(config-if)#no shut

Thu Sep 3 22:09:59 2026 Router1 AnupRajkafle2607985-R2(config-if)#exit

Thu Sep 3 22:10:10 2026 Router1 AnupRajkafle2607985-R2(config)#interface gig0/0/1

Thu Sep 3 22:10:27 2026 Router1 AnupRajkafle2607985-R2(config-if)#ip address 192.168.25.1 255.255.255.248

Thu Sep 3 22:10:31 2026 Router1 AnupRajkafle2607985-R2(config-if)#no shut

Thu Sep 3 22:10:36 2026 Router1 AnupRajkafle2607985-R2(config-if)#exit

Thu Sep 3 22:11:46 2026 Router1 AnupRajkafle2607985-R2(config)#interface se0/2/0

Thu Sep 3 22:12:16 2026 Router1 AnupRajkafle2607985-R2(config-if)#ip address 192.168.100.3 255.255.255.252

Thu Sep 3 22:13:04 2026 Router1 AnupRajkafle2607985-R2(config-if)#ip address 192.168.100.5 255.255.255.252

Thu Sep 3 22:13:08 2026 Router1 AnupRajkafle2607985-R2(config-if)#no shut

Thu Sep 3 22:13:12 2026 Router1 AnupRajkafle2607985-R2(config-if)#exit

Thu Sep 3 22:56:51 2026 Router2 AnupRajkafle2607985-R3>en

Thu Sep 3 22:57:06 2026 Router2 AnupRajkafle2607985-R3#config t

Thu Sep 3 22:57:50 2026 Router2 AnupRajkafle2607985-R3(config)#interface se0/2/1

Thu Sep 3 22:58:23 2026 Router2 AnupRajkafle2607985-R3(config-if)#ip address 192.168.100.6 255.255.255.252

Thu Sep 3 22:58:25 2026 Router2 AnupRajkafle2607985-R3(config-if)#no shut

Thu Sep 3 22:59:16 2026 Router2 AnupRajkafle2607985-R3(config-if)#exit

Thu Sep 3 22:59:30 2026 Router2 AnupRajkafle2607985-R3(config)#interface gig0/0/0

Thu Sep 3 23:00:12 2026 Router2 AnupRajkafle2607985-R3(config-if)#ip address 192.168.26.1 255.255.255.248

Thu Sep 3 23:00:15 2026 Router2 AnupRajkafle2607985-R3(config-if)#no shut

Thu Sep 3 23:00:18 2026 Router2 AnupRajkafle2607985-R3(config-if)#exit

Thu Sep 3 23:00:32 2026 Router2 AnupRajkafle2607985-R3(config)#interface gig0/0/1

Thu Sep 3 23:00:40 2026 Router2 AnupRajkafle2607985-R3(config-if)#ip address 192.168.27.1 255.255.255.248

Thu Sep 3 23:00:42 2026 Router2 AnupRajkafle2607985-R3(config-if)#no shut

Thu Sep 3 23:22:18 2026 Router0 AnupRajkafle2607985-R1>en

Thu Sep 3 23:28:21 2026 Router0 AnupRajkafle2607985-R1#config t

Thu Sep 3 23:29:13 2026 Router0 AnupRajkafle2607985-R1(config)#router ospf 1

Thu Sep 3 23:30:04 2026 Router0 AnupRajkafle2607985-R1(config-router)#network 192.168.22.0 0.0.0.7 area 0

Thu Sep 3 23:30:28 2026 Router0 AnupRajkafle2607985-R1(config-router)#network 192.168.23.0 0.0.0.7 area 0

Thu Sep 3 23:31:44 2026 Router0 AnupRajkafle2607985-R1(config-router)#network 192.168.100.0 0.0.0.3 area 0

Thu Sep 3 23:31:58 2026 Router0 AnupRajkafle2607985-R1(config-router)#no shut

Thu Sep 3 23:32:26 2026 Router0 AnupRajkafle2607985-R1(config-router)#show ip route ospf

Thu Sep 3 23:34:19 2026 Router0 AnupRajkafle2607985-R1(config-router)#do write memory

Thu Sep 3 23:37:01 2026 Router0 AnupRajkafle2607985-R1(config-router)#exit

Thu Sep 3 23:37:04 2026 Router0 AnupRajkafle2607985-R1(config)#show ip route ospf

Thu Sep 3 23:38:01 2026 Router0 AnupRajkafle2607985-R1(config)#show ip route ospf

Thu Sep 3 23:38:07 2026 Router0 AnupRajkafle2607985-R1(config)#show ip ospf neighbor

Thu Sep 3 23:38:20 2026 Router0 AnupRajkafle2607985-R1(config)#router ospf 1

Thu Sep 3 23:38:23 2026 Router0 AnupRajkafle2607985-R1(config-router)#show ip ospf neighbor

Thu Sep 3 23:39:19 2026 Router0 AnupRajkafle2607985-R1(config-router)#exit

Thu Sep 3 23:39:22 2026 Router0 AnupRajkafle2607985-R1(config)#exit

Thu Sep 3 23:39:41 2026 Router0 AnupRajkafle2607985-R1#show ip route ospf

Thu Sep 3 23:39:58 2026 Router0 AnupRajkafle2607985-R1#show ip ospf neighbor

Thu Sep 3 23:40:08 2026 Router1 AnupRajkafle2607985-R2>en

Thu Sep 3 23:40:12 2026 Router1 AnupRajkafle2607985-R2#config t

Thu Sep 3 23:40:33 2026 Router1 AnupRajkafle2607985-R2(config)#router ospf 1

Thu Sep 3 23:41:17 2026 Router1 AnupRajkafle2607985-R2(config-router)#network 192.168.100.0 0.0.0.3 area 0

Thu Sep 3 23:41:24 2026 Router1 AnupRajkafle2607985-R2(config-router)#network 192.168.100.4 0.0.0.3 area 0

Thu Sep 3 23:42:05 2026 Router1 AnupRajkafle2607985-R2(config-router)#network 192.168.24.0 0.0.0.7 area 0

Thu Sep 3 23:42:12 2026 Router1 AnupRajkafle2607985-R2(config-router)#network 192.168.25.0 0.0.0.7 area 0

Thu Sep 3 23:42:29 2026 Router1 AnupRajkafle2607985-R2(config-router)#do write memory

Thu Sep 3 23:42:59 2026 Router2 AnupRajkafle2607985-R3>en

Thu Sep 3 23:43:02 2026 Router2 AnupRajkafle2607985-R3#config t

Thu Sep 3 23:43:12 2026 Router2 AnupRajkafle2607985-R3(config)#router ospf 1

Thu Sep 3 23:43:43 2026 Router2 AnupRajkafle2607985-R3(config-router)#network 192.168.100.4 0.0.0.3 area 0

Thu Sep 3 23:43:55 2026 Router2 AnupRajkafle2607985-R3(config-router)#network 192.168.100.4 0.0.0.3 area 0

Thu Sep 3 23:44:24 2026 Router2 AnupRajkafle2607985-R3(config-router)#network 192.168.26.0 0.0.0.7 area 0

Thu Sep 3 23:44:33 2026 Router2 AnupRajkafle2607985-R3(config-router)#network 192.168.27.0 0.0.0.7 area 0

Thu Sep 3 23:44:42 2026 Router2 AnupRajkafle2607985-R3(config-router)#do write memory

Thu Sep 3 23:44:58 2026 Router2 AnupRajkafle2607985-R3#show ip route ospf

Thu Sep 3 23:45:36 2026 Router2 AnupRajkafle2607985-R3#show ip ospf neighbour

Thu Sep 3 23:45:57 2026 Router2 AnupRajkafle2607985-R3#show ip ospf neighbor

Fri Sep 4 14:24:11 2026 Router1 AnupRajkafle2607985-R2>en

Fri Sep 4 14:24:48 2026 Router1 AnupRajkafle2607985-R2#show ip route ospf

Fri Sep 4 14:30:33 2026 Router1 AnupRajkafle2607985-R2#show ip ospf neighbor

Sat Sep 5 22:23:04 2026 Router0 AnupRajkafle2607985-R1>en

Sat Sep 5 22:23:21 2026 Router0 AnupRajkafle2607985-R1#show ip route ospf

Sat Sep 5 22:23:40 2026 Router0 AnupRajkafle2607985-R1#show route ospf neighbour

Sat Sep 5 22:25:05 2026 Router0 AnupRajkafle2607985-R1#show ip ospf neighbor

Sat Sep 5 22:26:17 2026 Router1 AnupRajkafle2607985-R2>en

Sat Sep 5 22:26:28 2026 Router1 AnupRajkafle2607985-R2#show ip route ospf

Sat Sep 5 22:27:00 2026 Router1 AnupRajkafle2607985-R2#show ip ospf neighbor

Sat Sep 5 22:28:22 2026 Router0 AnupRajkafle2607985-R1#show ip route ospf

Sat Sep 5 22:28:24 2026 Router0 AnupRajkafle2607985-R1#show ip ospf neighbor

Sat Sep 5 22:29:53 2026 Router2 AnupRajkafle2607985-R3>en

Sat Sep 5 22:30:02 2026 Router2 AnupRajkafle2607985-R3#show ip route ospf

Sat Sep 5 22:30:24 2026 Router2 AnupRajkafle2607985-R3#show ip ospf neighbor